

Role of thermal fluid dynamics in alloying element distribution and weld porosity in powder feeding-based laser welding of Al alloy

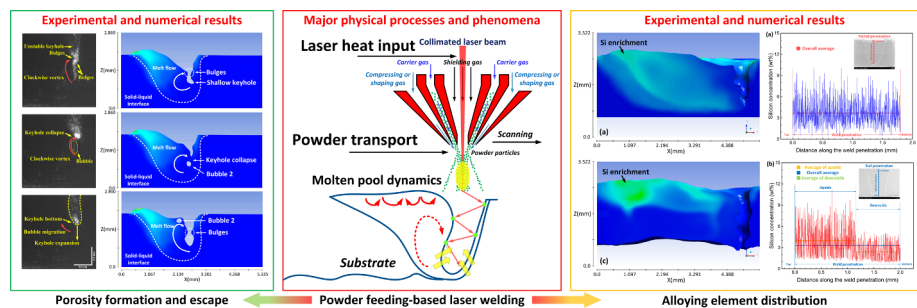
Zhenglong Lei, Shibo Wu*, Bingqi Cao, Bingwei Li, Jingwei Liang, Xue Hu, Yanbin Chen

State Key Laboratory of Advanced Welding and Joining, Harbin Institute of Technology, Harbin 150001, China

HIGHLIGHTS

- A 3D model incorporating the powder transport, laser heating, and molten pool dynamics in an integrated manner is developed.
- Effects of different melt flow behaviors on the distribution of alloying element are investigated.
- Three necessary steps for bubble formation and migration are concluded.
- A Glass-Al composite sample is designed to observe keyhole and molten pool behaviors.
- The obtained results are of great significance to providing a guidance for process optimization.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Powder feeding-based laser welding
Thermal fluid dynamics
Melt flow
Alloying element distribution
Weld porosity

ABSTRACT

This work develops a model incorporating the powder transport, laser heating, and molten pool dynamics to describe the mixing process of alloying element and reveal the mechanisms of bubble dynamics in powder feeding-based laser welding. Simulation results show that the different roles of recoil pressure: driving the molten metal flow at the rear part of keyhole in partial penetration and maintaining the stable keyhole profile in full penetration, are responsible for the difference between their melt flow behaviors. Driven by the melt flow, the alloying element transport path can be generalized as the shallow areas, the rear areas, the central and bottom areas in partial penetration. It permits a relatively uniform silicon distribution. However, two vortices rotating in the opposite direction in full penetration results in a higher concentration of Si in the upper part. Meanwhile, we conclude three steps for the bubble dynamics: bulge formation, bubble formation, bubble being captured by re-formed keyhole or being captured by the solid-liquid interface or escaping from the free surface. Such phenomena manifest that the contributions to the processing quality and metallurgy quality of welded joints come from the complex melt flow behaviors, which are confirmed by a series of experiments.

1. Introduction

Aluminum alloys are extensively used in lightweight structures in commercial, industrial and military fields [1,2]. Laser beam welding technology (LBM), which is usually preferred over conventional

welding processes due to low heat input, high energy density and high welding speed [3], has been implemented across Al alloys manufactured products such as thick plates in shipbuilding and thin plates in automobile industry [4,5]. Despite the successes, the rapid development of Al alloys laser welding is limited due to a series of problems,

* Corresponding author.

E-mail address: wushibohit@163.com (S. Wu).

<https://doi.org/10.1016/j.applthermaleng.2020.115081>

Received 17 October 2019; Received in revised form 19 January 2020; Accepted 12 February 2020

Available online 13 February 2020

1359-4311/ © 2020 Elsevier Ltd. All rights reserved.

Nomenclature

C_m	drag coefficient
d	average bubble diameter
F_B	buoyancy
F_D	drag force
F_G	gravitational force
Re	Reynolds

S bubble migration distance

Greek symbols

ψ	alloying element weight percentage
ρ	density
μ	molecular viscosity
θ	drag angle

including the high reflectivity which results in most laser energy being reflected from the material surface [6], the weld porosity induced by keyhole instability or metallurgical factors [7], and the poor weldability of dissimilar materials [8–10].

Powder feeding-based laser beam welding technology (PFLBW) has been applied initially in some areas of machinery as a combined method of laser welding and laser melting deposition; it provides the possibility of improving the alloy composition of welded joints and joining dissimilar materials [11–13]. Powder particles carried by the gas flow through a nozzle are transported to the molten pool formed by the laser heat source. Although the addition of powder was proven to help reduce the attenuation of laser [14], accelerate the heat transfer in laser welding [15], and modify the microstructure of the welded joints [16,17], which is still considered as an instability factor during laser welding. To obtain a wide welding process window, Katayama et al. [18–20] employed the high speed X-radiography as a process monitoring approach to investigate the process characteristics of laser welding including keyhole profile and molten pool behaviors. However, specialized equipment and materials involved in the experiments are costly. Numerical modeling as a visual approach has been carried out to be valuable to understanding thermodynamics behaviors in PFLBW [21].

With the development of numerical simulation studies, considerable attention should be paid on the improvement of practical models, contributing to a comprehensive analysis of calculation procedure and full consideration of physical phenomena. In other words, complex physical phenomena involved in PFLBW require to be formulated accurately in the simulation. In recent years, many scholars have spent a

lot of effort to solve related problems. Existing studies on modeling the interaction between laser and materials are given through two aspects:

In the first part, some works concerning powder material deposition, which pay more attention to the gas-dispersed flows, powder transport, laser-induced heating, and continuous bead deposition, are considered. Pan et al. [22] and Arrizubieta et al. [23] used numerical process simulations to analyze the transport characteristics of powder particles supplied with the triple coaxial nozzle, the effects of nozzle geometry were addressed as well. Nie et al. [24] performed the modeling of laser cladding in three sub-steps to study the impact of processing parameters on the thermal history, which further affected and controlled the deposited coating microstructure. But the interaction of the particle mass-flow and molten pool was beyond their consideration. Kovalev et al. [25,26] modeled the coaxial laser cladding based on a 3D description of the gas-disperse jet flows, powder transportation, laser heating, and bead surface growth, but the effects of process parameters on the molten pool flow behaviors are not considered. Wen and Shin [27] presented a 3D transient model for a coaxial laser direct deposition process in which they calculated the evolution of the free interface. However, the mechanism of the surface build-up has been simplified.

In the second part, only thermal fluid dynamics, which are responsible for heat and mass transfer, melt flow, and quality evolution, are simulated. Ki's group [28,29] established a 3D model involving multiple reflections for reproducing the keyhole formation. The keyhole free surface evolution was also tracked through the level set method. Panwisawas et al. [5] developed a thermal fluid dynamics model allowing for the phase transition of the metallic component to capture the transient physical effects in laser welding of titanium alloys.

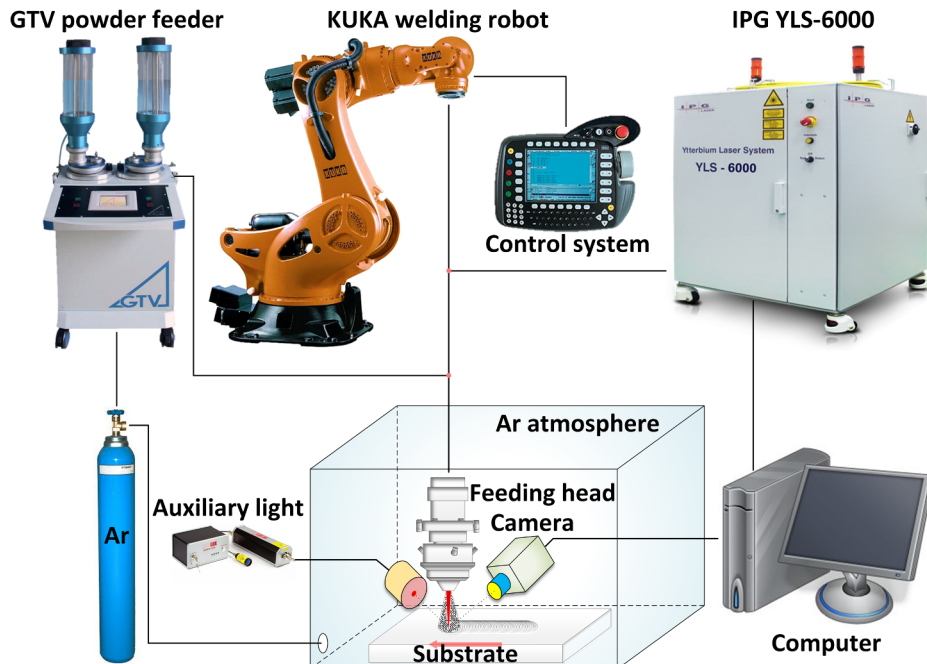


Fig. 1. Schematic diagram of the powder feeding-based laser welding platform, equipped with an IPG YLR 6000 fiber laser, a GTV PF 2/2 coaxial powder feeder, a KUKA welding robot, an Optronics CR 5000 high-speed camera system and a visible argon gas chamber.

Table 1
Process parameters for powder feeding-based laser welding.

Process No.	Laser power(kW)	Welding speed (m/min)	Average Powder impact velocity (m/s)	Powder temperature (K)
1	3.5	8	2.0	1850
2	4.0	8	2.0	1850

With the development of the research into the laser-material interaction, there exists considerable concern for modeling periodic keyhole behaviors through the VOF method [30]. Zhao et al. [31] proposed a three-phase coupled model in which they summarized that the keyhole collapse arising from the strong melt flow, together with shrinkage, was responsible for porosity formation. Huang et al. [32] and Lin et al. [33] investigated the keyhole instability through a comparative study of molten pool dynamics, and they also summarized three necessary steps for keyhole collapse and bubble formation in LBM. However, the mechanism of bubble escaping has been significantly simplified. Available numerical approaches by Cho and Na's group [34–36] permitted not only predicting the porosity distribution but also describing the alloying element mixing process in laser-GMA hybrid welding, which were extended by Muhammad et al. [37] to study the effects of GMA torch orientation on the mixing behaviors of molten pool. Esfahani et al. [38] performed the modeling of dissimilar laser welding to calculate the dilution and homogeneity of the welded joint. However, the relationship between molten pool dynamics and alloy composition was beyond the authors' consideration.

Besides the processes of laser welding, Masoomi et al. [39] revealed process-property relationships under a varying number of lasers and scan strategies via thermal modeling of the laser powder bed fusion process. Xia et al. [40] calculated molten pool dynamics during selective laser melting (SLM) via the finite volume method. They also demonstrated an appropriate hatch spacing contributed to spreading the molten liquid and obtaining a flat surface with low roughness. Their SLM physical model is later extended by coupling a discrete element method model by Gu et al. [41] to study the effects of powder packing characteristics on the flow behaviors.

Despite much research reported to understand the complicated physical phenomena involved in the laser-material interaction, the interest for modeling PFLBW is to predict the process stability of heat and mass transfer under the various strategies and materials, and then to control the microstructure and enhance the performance of the welded joint.

Based on the previous research [21], a 3D model incorporating the powder transport, laser heating, and molten pool dynamics in an integrated manner is proposed to yield a practically reasonable solution of PFLBW problems. The relationship of thermal fluid dynamics to alloying element distribution and weld porosity are also discussed. The experimental characterizations of alloying element distribution of the welded joints and the experimental observation of dynamic molten pool and keyhole behaviors are performed to validate the results obtained by numerical simulations.

2. Experimental procedure and modeling approach

2.1. Experimental material and procedures

To reveal the relationship of thermal fluid dynamics to alloying element distribution and weld porosity in PFLBW, a 6 kW CW fiber laser

with the wavelength of 1.07 μm (IPG YLR-6000, Germany), and a coaxial powder feeder (GTV PF 2/2, Germany) are combined to conduct single-path 2060 Al alloys overlaying welding, as shown in Fig. 1. In the process, the focus laser beam acts on the top surface of the substrate and the focus radius is measured as about 0.6 mm. The chemical compositions in weight percentage of 2060 aluminum alloy are shown in Table 2.

Before welding, the substrate surface is cleaned by a steel brush and acetone. To decrease the weld porosity, the key process parameters are tested and optimized at a large range. Table 1 lists the process parameters used in our cases (i.e., No.1 corresponding to the partial penetration welding and No.2 corresponding to the full penetration welding). During welding, a high-speed CCD camera (Optronis CR 5000, Germany) is applied to capture molten pool characteristics at 5000 fps, and a laser-assisted light source (CAVILUX HF, Finland) with a wavelength of 810 nm and maximum power of 500 W is selected to illuminate the keyhole and molten pool. After welding, the metallurgical samples are mechanically cut perpendicular to the center line of weld and ground with silicon carbide paper. To obtain a mirror-like surface quality, subsequent electrolytic polishing is performed to remove the deformed surface layer resulting from the mechanical processing. The alloying element distribution of the welded joints after etched in 2 vol% HF solution is characterized by Energy Dispersive Spectrum (Quanta 200FEG-SEM, FEI Company, USA). The detailed analysis results are described in the sections below.

2.2. Physical phenomena and general assumptions

According to our view, simulations provide a visual approach to understanding the dynamic physical processes of PFLBW in an integrated manner. Fig. 2 summarizes the major physical phenomena involved in PFLBW, including laser heating, powder transport, phase changes, molten pool dynamics and plasma. To establish the physically consistent mathematical model, we must introduce some basic assumptions for simplification while preserving the effect of the main factors governing the process. Based on the considerable literature, the basic assumptions are listed as follows:

- (1) Considering that gas-atomized powder particles are almost spherical and fully dense, the powder particles of spherical shape (300 μm in diameter) with the predefined distribution function are used to improve computing efficiency [21];
- (2) Considering the complicated chemical components of the powder mixture, it is a general way in simulation to treat the mixed powder of 2060 Al alloy (50%) and Si (50%) as a continuum medium with an average density, heat capacity, and thermal conductivity to be transported into the molten pool [25];
- (3) Considering the contributions to the powder transport from the drag force are far more than those from the sum of other hydrodynamic forces, both the interaction between particles and nozzle and the collision between powder particles are not considered [26];

Table 2
Chemical composition of 2060 aluminum alloy used in the simulation.

Element	Cu	Li	Zn	Mg	Mn	Ag	Zr	Ti	Al
(wt%)	3.9	0.8	0.32	0.7	0.29	0.34	0.1	< 0.1	Balance

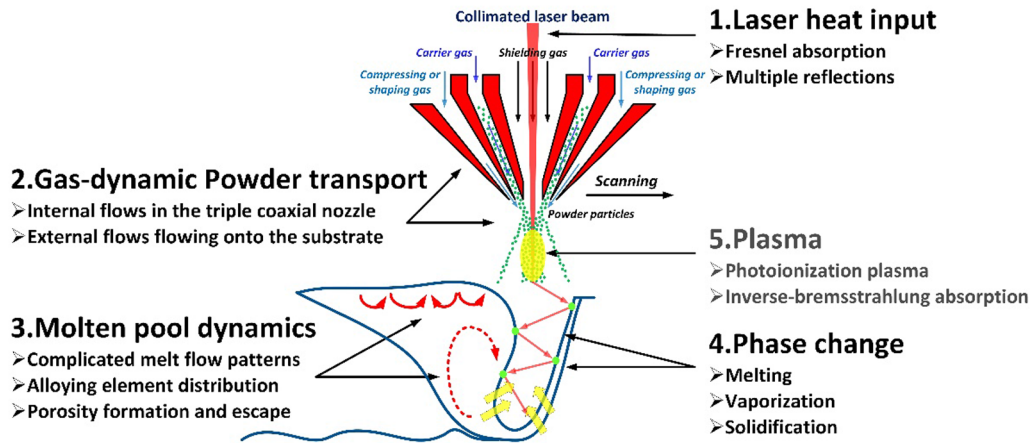


Fig. 2. Major physical processes and phenomena in powder feeding-based laser welding, including laser heat input, gas-dynamic powder transport, phase changes, molten pool dynamics, and plasma.

- (4) Under the conditions of the coaxial supply of the powder (metals and ceramics) during the laser manufacturing, much attention is paid to the effects of powder particles on the molten pool. Laser shielding gas jet and frictional force of metal vapor are assumed to exert no effects on molten pool dynamics [42,43];
- (5) As is known [33], in practice, it is preferable to regard the molten metal as an incompressible Newtonian fluid in simulation of molten pool dynamics.

2.3. Modelling of powder feeding-based laser welding

Note that the details of the mathematical models, Navier-Stokes equations and other basic equations of powder transport are referenced to [21], and we do not repeat them here. To describe the alloying element distribution in a cell, the additional conservation equation is given by:

$$\frac{\partial \psi}{\partial t} + \nabla \cdot (\vec{v} \psi) = 0 \quad (1)$$

Here, ψ as a scalar quantity denotes the weight percentage of alloying element in a cell, which is dominated by molten pool dynamics. However, the diffusive effects of the alloying element are beyond consideration in this case. Fig. 3 shows the computational domain and the mesh used in the process simulation. To balance the computational efficiency and the computational cost, the adaptive mesh refinement method is used for the effective mesh transition from the weld center-line to the edge, and the minimum mesh size is designed as 0.1 mm. Considering the geometric symmetry, half of the computational domain consisting of 77,870 elements and 85,529 nodes is adopted in the

simulation. The thermos-physical material properties used for fluid flow and heat transfer calculations are supported by Lei et al. [21] and Han et al. [44,45].

In our model, to suppress the numerical parasitic flow, the balanced forced continuum surface force algorithm (CSF) [46–48] is applied to accurately deal with the boundary condition of the molten pool surface. The details of the body force model [5] involved in the CSF algorithm and the energy balance equations have been addressed in our previous study [21]. To effectively ensure the convergence of the computing process taking into account both the laser beam position and the continuous powder feeding, the boundary condition settings at the region of the feeding powder inlet are supposed to be the Dirichlet type. At the other region of the computational domain, the boundary conditions are set as the Neumann type.

2.4. Numerical calculation procedure

To provide modeling consistency between the totality of physical processes, the basic calculation procedure is performed to model dynamic melt flow behaviors, and their effects on the alloying distribution and weld porosity during PFLBW. The flow chart of calculation procedure is shown in Fig. 4.

3. Results and discussion

3.1. Thermodynamic behavior of molten pool

Thermodynamic behaviors of molten pool determined by the driving forces are responsible for transporting the alloy element to a

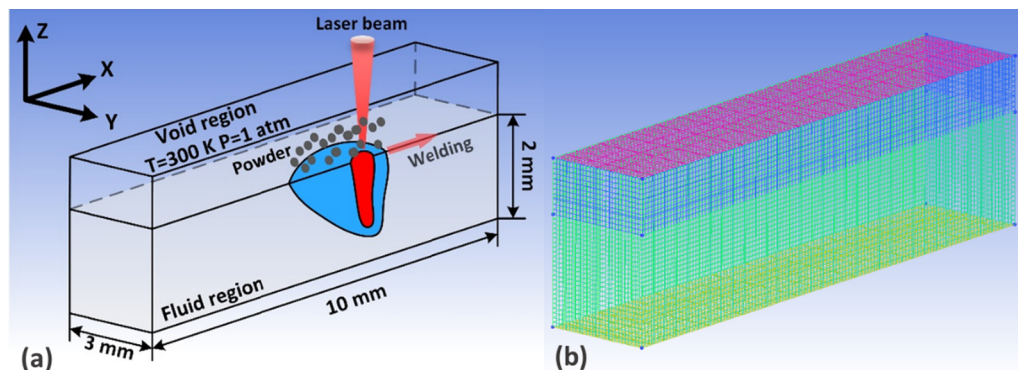


Fig. 3. Illustration of the computational domain and the mesh used in the simulation: (a) half of the computational domain (a fluid region of 10 mm × 3 mm × 2 mm and a void region on top of the fluid region for tracking free surface deformation) and (b) grid division result.

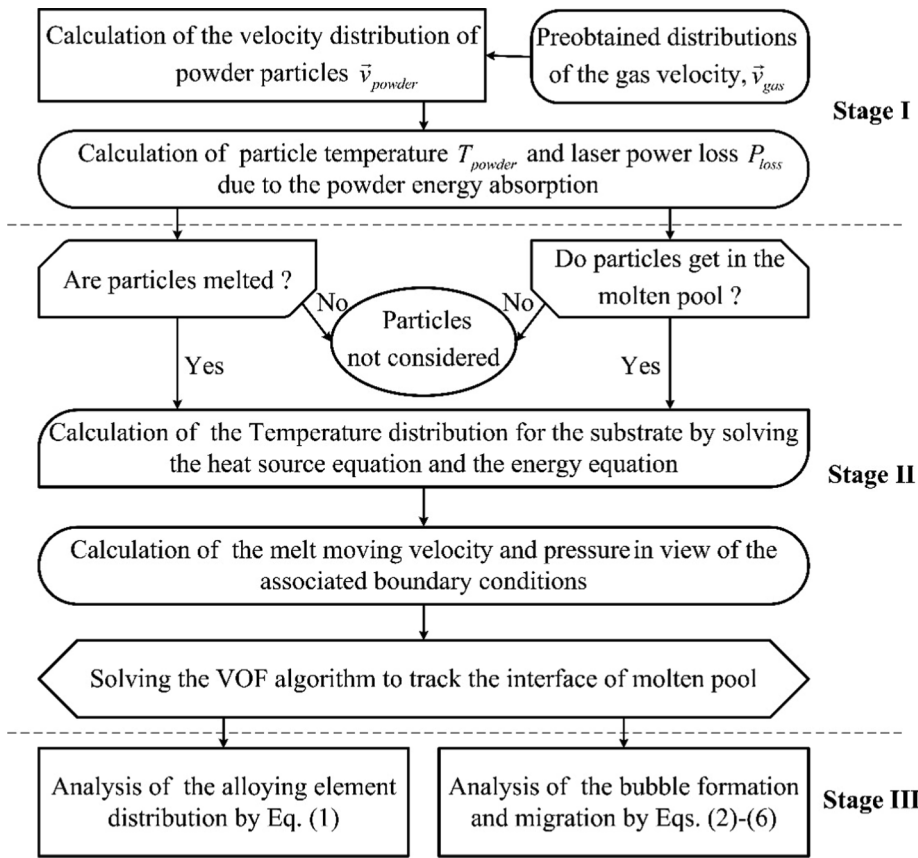


Fig. 4. Basic calculation procedure of processes accompanying the powder feeding-based welding (Stage I: based on the known gas velocity distribution of Trumpf DMD505 [25,26], the discrete-trajectory method is used to solve the velocity, the mass flow and the temperature distribution of powder particles; Stage II: first, the continuity equation, momentum equations and energy equations are solved to obtain the velocity, pressure and temperature distributions in consideration of the associated boundary conditions. Secondly, the VOF algorithm is solved to track the interface of molten pool; Stage III: the additional alloying element conservation equation and the bubble migration model are solved to analyze the alloying element distribution, and the bubble migration).

wide range of molten pool and inducing porosity formation. As described in the model, the main driving forces are composed of the recoil pressure, surface tension, and the powder impact force. Considering the multiple laser reflections occurring on the keyhole walls with irregular profile, the reflected beam can reach different locations of keyhole walls and can be absorbed. The resulting temperature variation determines the distribution of the recoil pressure and surface tension. That is, the increase of surface temperature leads to the higher recoil pressure and the lower surface tension, and hence the molten metal flows much more strongly in the high temperature region. Also, the powder

impact force, which is related to the powder falling velocity, has been proven to only act on the shallow areas of molten pool. That is because the high-viscosity and density molten metal opposes the downward motion of powder particles [21].

Fig. 5 shows the calculated melt flow patterns in the longitudinal sections in PFLBW. With regard to the partial penetration welding in Fig. 5(a), a thin molten layer due to high-intensity laser interaction with metals is found in front of the keyhole, where a random melt flow can be captured as well. Meanwhile, several complex melt flow patterns exist at the rear part of the keyhole. For example, there are a number of

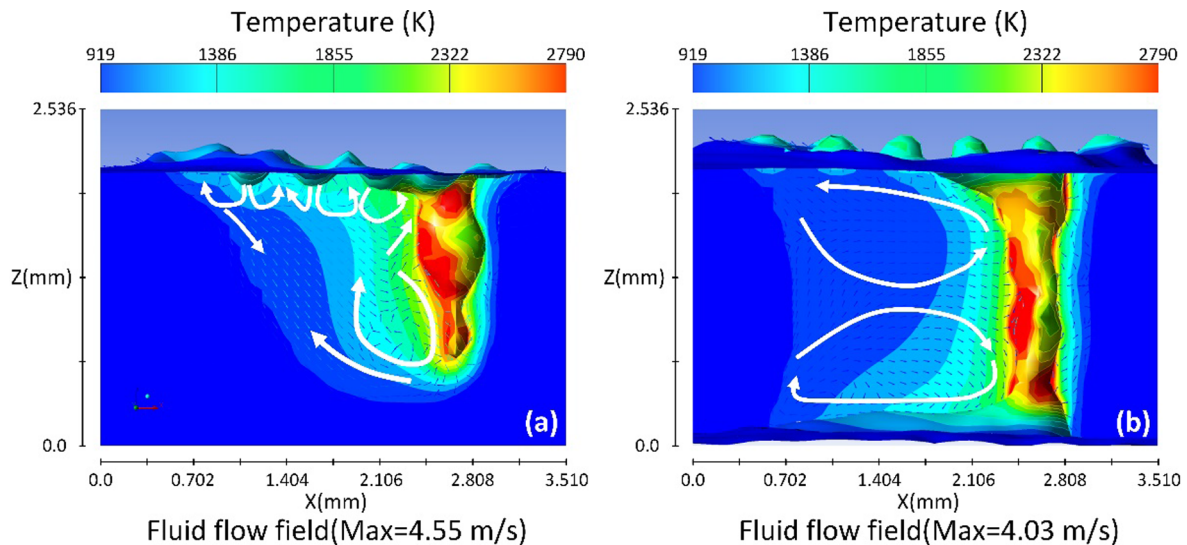


Fig. 5. Calculated melt flow patterns in the longitudinal sections during powder feeding-based laser welding: (a) partial penetration (Process No.1) and (b) full penetration (Process No.2).

ripples and pits on the upper part of molten pool, which are caused by the falling powder particles with relatively high velocity. As a result, plenty of small vortices are formed in the upper parts of molten pool, which are attributed to the downward impact force on the molten metal located at landing positions and the upward extrusion force on the molten metal between adjacent landing positions. Considering that the reflected beam can reach the different position of the rear keyhole wall, the rear part of the keyhole absorbs large quantities laser heat. Consequently, the molten metal at the upper part gets a downward flow along with the keyhole wall due to the recoil pressure, and a rear upward flow caused by the impact of the incoming liquid from the bottom of keyhole is found at the rear part of keyhole. Such phenomena permit a clockwise vortex to transport the molten metal to other region and then to extend the area of molten pool. When it comes to the full

penetration welding in Fig. 5(b), the recoil pressure plays an important part in maintaining stable keyhole profile rather than driving the melt flow at the rear part of keyhole. Two vortices in the opposite direction are observed at the upper and lower parts of molten pool. Such a flow trend is caused by the Marangoni effect, which is associated with surface tension gradient [33].

3.2. Mixing behavior and distribution of alloying element

To highlight the influence of melt flow on the alloying element distribution, we select silicon as the tracer element to visualize the alloying element mixing process in the molten pool during PFLBW, which is attributed to its almost zero concentration in the base metal while 50% concentration in the case of filler powder. Fig. 6 shows the mixing

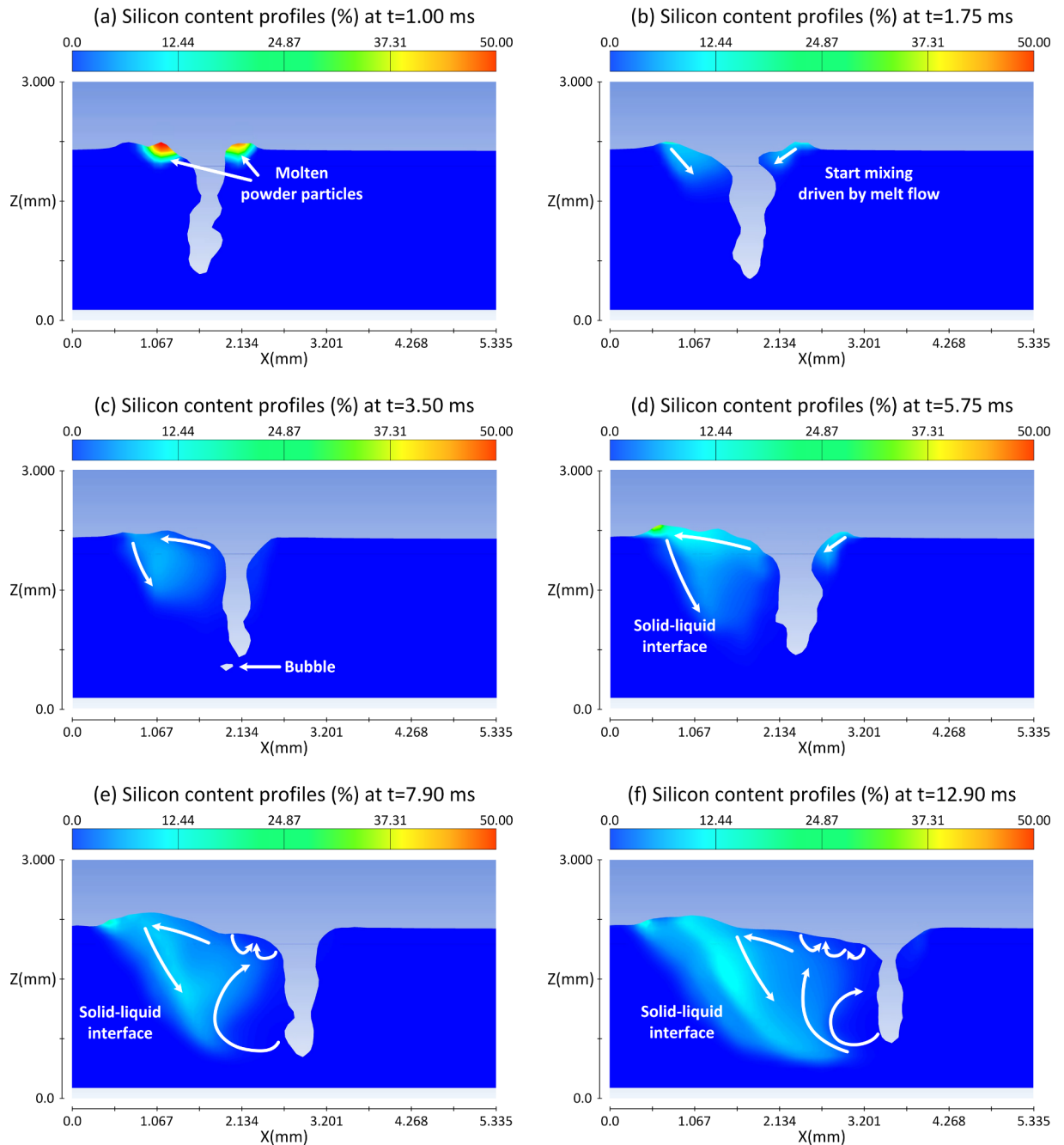


Fig. 6. Calculated Si transport process in longitudinal cross sections during powder feeding-based laser welding: silicon content profiles (%) at (a) 1.00 ms, (b) 1.75 ms (c) 3.50 ms, (d) 5.75 ms, (e) 7.90 ms and (f) 12.90 ms.

process of Si driven by melt flow in the longitudinal sections during partial penetration PFLBW.

In the beginning stages, plenty of molten powders particles with relatively high Si content fall vertically and make contact with the molten pool surface, which also permits Si to enter into the molten pool. As aforementioned, the filler powder makes a difference to the molten pool behaviors in the shallow areas, which also has a significant influence on the Si distribution. To be specific, the Si enrichment in the shallow areas is observed in Fig. 6(b) due to the shallow molten pool and inadequate melt flow at the very onset, and then Si is transported to the rear upper part of molten pool driven by surface tension. Meanwhile, the downward flow with high Si content around the solid-liquid interface helps transport Si from the rear upper part to the bottom of molten pool. When it comes to Fig. 6(e), the clockwise vortex at the rear part away from the keyhole plays a crucial role in transporting Si to the central area and a wide range of molten pool, which contributes to the uniform distribution of Si in the molten pool. Fig. 6(f) shows the low content of Si in the shallow areas close to the keyhole and in the rear part of the rear keyhole wall, which is caused by the strong melt flow driven by the powder impact force and the recoil pressure. It is worth noting that there also is a low content of Si in front of the forward keyhole wall in spite of the addition of powder. On the one hand, the melt flow has little effect on transporting Si to other regions due to the narrow melting area, and the downward transfer of Si under the stronger random flow trend is almost inhibited in this case. On the other hand, it can be found that the molten metal in front of the forward keyhole wall tends to flow round the keyhole to the rear part of molten pool, which contributes to distributing Si uniformly. In summary, the alloying element transport path in the molten pool during PFLBW (see Fig. 7(b)) can be generalized as the shallow areas, the rear areas, the central and bottom areas of molten pool, which distinguishes itself from that during laser-GMA hybrid welding [35,37].

As aforementioned, laser power not only has a vital influence on

laser welding modes but also results in different weld penetration, which makes a difference to the melt flow patterns and alloying element distributions. Fig. 7 shows the calculated Si distributions in the longitudinal sections under partial penetration and full penetration. With regard to partial penetration welding (see Fig. 7(a)), Si has a uniform distribution in the wide molten pool. Meanwhile, the enrichment of Si can be captured in the rear upper part of molten pool, which can be explained by the fact that high Si content is mostly circulating in the rearward melt flow at the upper part and quickly captured by the upcoming solidification front, considering that the molten pool is shallow at the beginning of welding. It is noteworthy that there is a low content of Si existing at the bottom of molten pool close to the solid-liquid interface, which is caused by the turbulent up-and-back flow related to the fast cooling rate and small accompanying molten zone [35]. As for full penetration welding (see Fig. 7(c)), the enrichment of Si in the rear upper part also can be observed. Considering the change in the melt flow patterns as addressed in Section 4.1, two vortices mainly driven by surface tension play a crucial role in transporting Si to the wide range of molten pool under full penetration welding. However, two vortices keep on rotating in the opposite direction, which causes trace amounts of Si to reach the bottom of molten pool. It permits a higher concentration of Si in the upper part of molten pool, which corresponds to the previous simulation results in [37].

3.3. Mechanisms of bubble formation and migration

As shown in Fig. 7, there is no weld porosity existing in the freezing molten pool with laser power of 3.5 kW and a high welding speed of 8 m/min. In this case, the bubbles have enough time to escape from the free surface. According to the previous study [21], the bubble formation is related to the keyhole instability, which is again correlated with molten pool dynamics. However, more attention has been paid for the effects of welding parameters on the depth oscillation amplitude and

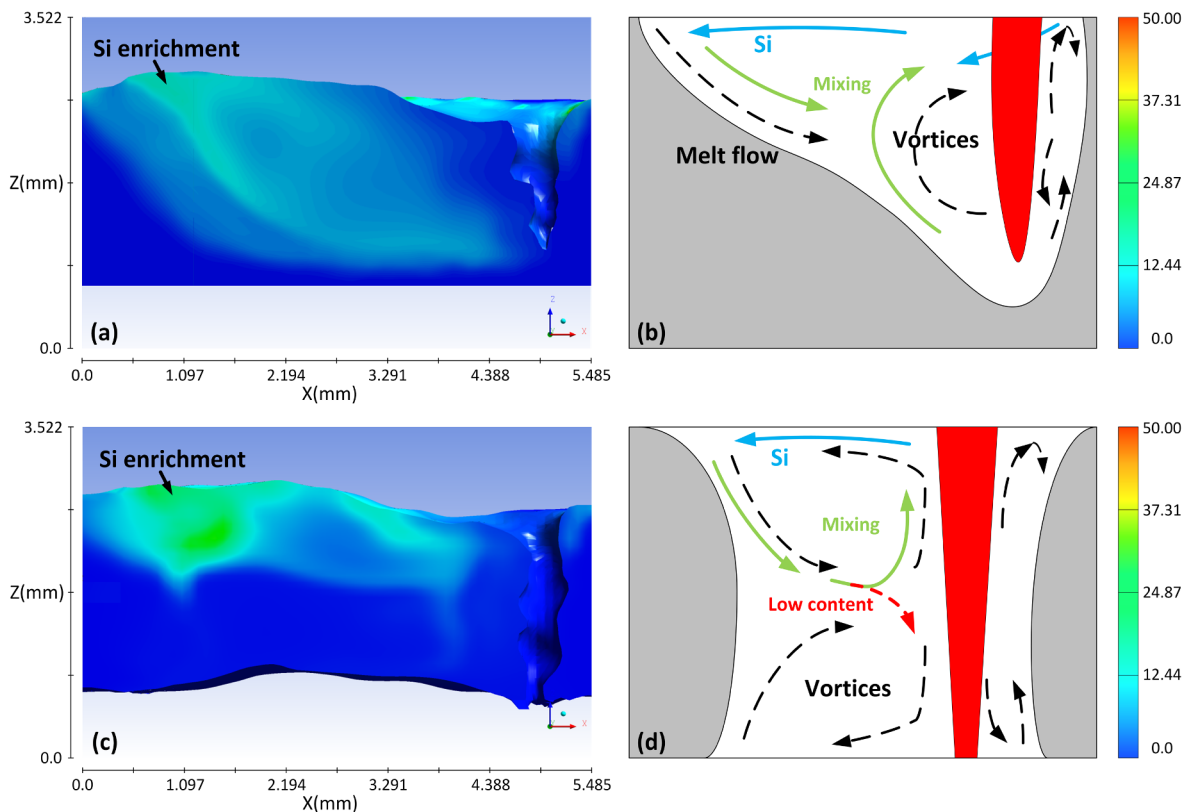


Fig. 7. Simulation results in the longitudinal sections (a) Si distributions under partial penetration (Process No.1), (b) schematic transport path of Si under partial penetration (Process No.1), (c) Si distributions under full penetration (Process No.2), and (d) schematic transport path of Si under full penetration (Process No.2).

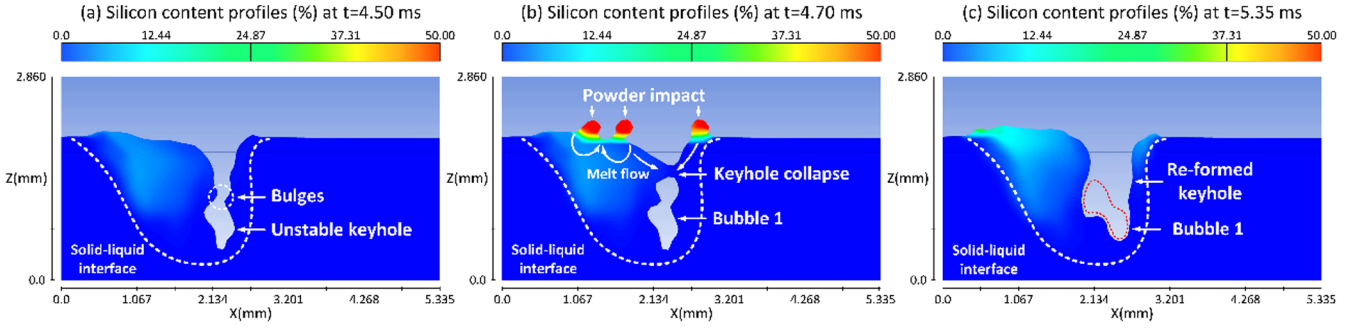


Fig. 8. The bubble migration behavior of being captured by re-formed keyhole: (a) 4.50 ms, (b) 4.70 ms and (c) 5.35 ms.

collapsing frequency of keyhole. In this section, the role of melt flow in bubble formation and migration is addressed as follows:

During PFLBW, the falling filler powder as an instability factor has the possibility to exacerbate the keyhole instability. In particular, the powder particles falling around the keyhole could lead to a decrease of keyhole entrance size and even the closing of keyhole. As shown in Fig. 8(a), driven by the powder impact, the molten metal in the shallow areas around the keyhole gets extruded to flow toward the keyhole wall, causing the bulges at the keyhole walls. Considering the large depth-to-width ratio of Al alloy and the strong melt flow in the shallow areas, in some cases, the bulges existing at the front and rear keyhole walls tend to make contact with each other, resulting in a keyhole collapse and a bubble in the bottom (see Fig. 8(b)). On the other hand, the clockwise-rotating vortex flow behind the keyhole plays an important part in promoting the melt flow and extending the area of molten pool. The strong vortex flow also permits the molten metal in the rear part of keyhole to flow toward the keyhole wall in some cases, resulting in the bulges at the rear keyhole wall (see Fig. 9(a)). The bulges continue to grow up and make contact with the front keyhole wall, as a result, a keyhole collapse and a bubble could be found in the bottom (see Fig. 9(b)). Such phenomena evidence that the contributions to the formation of bubbles in the bottom come from the keyhole collapse due to the strong melt flow, which is again driven by the powder impact force and recoil pressure.

It is worth noting that not all initial bubbles in the molten pool will turn into weld porosity in the welded joint. Much more attention should be paid to the force condition, migration and capture of bubbles. According to the previous study of Li et al. [49], the movement of a bubble was significantly affected by gravity (F_G), buoyancy force (F_B), and viscous drag force (F_D). In our model, the migration dynamics equations of an individual bubble can be described as follows:

$$F_{bubble} = -F_B + F_G + F_D = \frac{\rho_b \pi d^3}{6} \cdot \vec{a} \quad (2)$$

$$\frac{\partial \vec{v}_b}{\partial t} = \frac{\rho_b - \rho_m}{\rho_b} \vec{g} + C_m (\vec{v} - \vec{v}_b) + C_m (\vec{v} - \vec{v}_b) \cdot (\vec{v} - \vec{v}_b) \cdot \frac{\rho_m}{\rho_b} \quad (3)$$

$$C_m = \frac{3d}{4} \left(\frac{24}{Re} + \frac{6}{1 + \sqrt{Re}} + 0.4 \right) \quad (4)$$

where ρ_b is the gas density in the bubble, ρ_m is the density of molten metal, $\vec{v}$ is the melt flow velocity, $\vec{v}_b$ is the bubble moving velocity, d is the average bubble diameter, μ_m is the viscosity coefficient, and C_m is the drag coefficient between the bubbles and the melt. Considering that the Reynolds number of the fluid is associated with the viscosity coefficient ($Re = d \rho_m |\vec{v} - \vec{v}_b| / \mu_m$), the bubble migration model could be simplified as follows [50]:

$$\frac{\partial \vec{v}_b}{\partial t} = \frac{\rho_b - \rho_m}{\rho_b} \vec{g} + \frac{18 \mu_m \vec{v}_b}{d^2 \rho_b} \quad (5)$$

The bubble migration distance in the direction perpendicular to the welding direction could be written by:

$$S = v_b t \cos \theta + \frac{1}{2} \left(\frac{\rho_m - \rho_b}{\rho_b} \vec{g} - \frac{18 \mu_m v_b \cos \theta}{d^2 \rho_b} \right) t^2 \quad (6)$$

where t is the bubble migration time, and θ is the drag angle between the drag force and gravity. The direction of drag force is proven to be almost parallel to the outline of lower part of molten pool [50].

As above, the formation of bubbles results from the keyhole collapse. Meanwhile, the dimension characteristics of bubbles are closely correlated with the keyhole collapse position. Given that the keyhole collapse caused by the powder impact occurs at the middle-upper part of keyhole while that arisen from the recoil force could be found at the middle-lower part of keyhole, hence the bubbles generated by the former are generally much larger than those generated by the latter. As a result, there are a lot of differences between their bubble migration processes. Fig. 8(b) shows a bubble with a large size in the bottom of keyhole, which is named bubble1. Based on Eq. (6), it will take more time for the larger bubble to escape from the free surface at the same

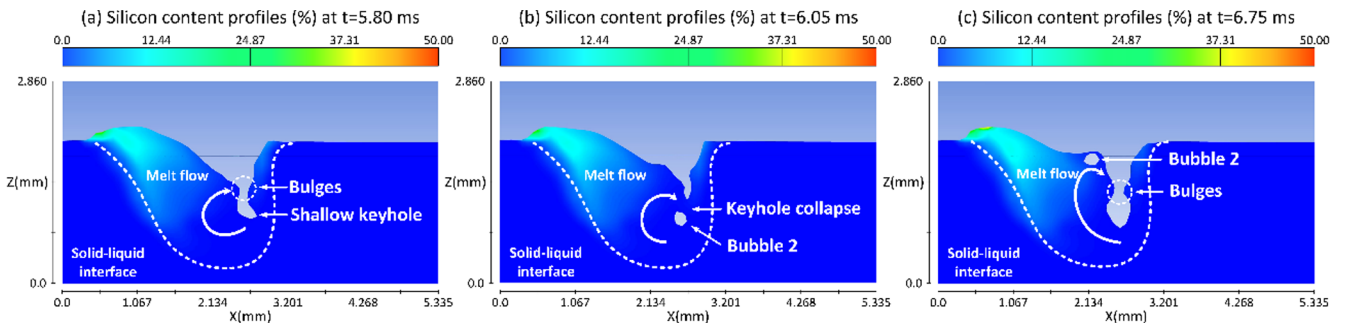


Fig. 9. The bubble migration process from the bottom to the surface of molten pool: (a) 5.80 ms, (b) 6.05 ms and (c) 6.75 ms.

vertical bubble migration distance. At the same time, the larger surface area of bubbles raises their possibility of being captured by the re-formed keyhole in the migration process. As shown in Fig. 8(c), bubble 1 has been captured successfully, causing no pore formation in the molten pool. When it comes to Fig. 9(b), bubble 2, caused by the keyhole collapse, is found in the bottom of keyhole as well. Meanwhile, the high recoil pressure generated in the keyhole re-formation process contributes to the strong vortex flow, which drives bubble 2 to flow upwards and backwards. As a result, bubble 2 succeeds in escaping from the free surface, causing no pore formation in the molten pool as well.

Such phenomena as stated above reveal that the keyhole collapse caused by the powder impact and the vortex flow is the main reason leading to the bubble formation in the bottom. Meanwhile, the molten pool dimensions and melt flow characteristics have great influence on the bubble migration. Fig. 10 summarizes the mechanisms of bubble formation and migration in three necessary steps: bulges being formed at the keyhole wall, bubble being formed due to the keyhole collapse, and bubble being captured by the re-formed keyhole or escaping from the free surface or being captured by the solid-liquid interface to become a pore.

Note that there is no bubble being captured by the solid-liquid interface in the case of high-speed welding ($P = 3.5$ kW and $v = 8$ m/min). Based on the previous study [32,33], the higher welding speed results in the greater keyhole entrance size (L_k) and molten pool length (L_M), which contribute to improving keyhole stability and providing more time for bubbles to float out. Compared to low-speed welding, the rear keyhole wall bears the smaller recoil pressure, leading to the smaller melt flow velocity (approximately equal to the bubble moving velocity, v_b) and the larger drag angle (θ). It permits not only the lower keyhole collapse frequency but also the shorter time (t) in the same

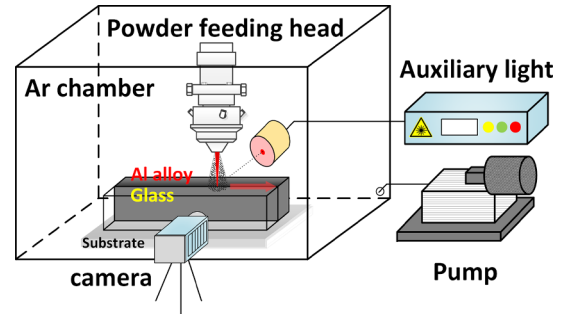


Fig. 11. Schematic of the Glass-Al composite sample and for observation of welding phenomena.

bubble migration vertical distance (S), which are greatly beneficial for preventing the bubble formation and promoting the bubble migration.

3.4. Experiment verification

According to the previous work during PFLBW [21], the agreement between simulation results and experimental results on the weld shapes, typical melt flow patterns and effects of powder on thermal fluid dynamics has been carried out to validate the physico-mathematical model. In this section, much more attention has been paid for the experiment verification of alloying element distribution, dynamic molten pool and keyhole behaviors obtained in the simulation.

At first, we not only set up an observation window equipped a high-speed camera system at specific locations, but also design a Glass-Al composite sample (see Fig. 11) to conduct a real-time monitoring for dynamic keyhole behaviors during PFLBW, as displayed in Fig. 12. The

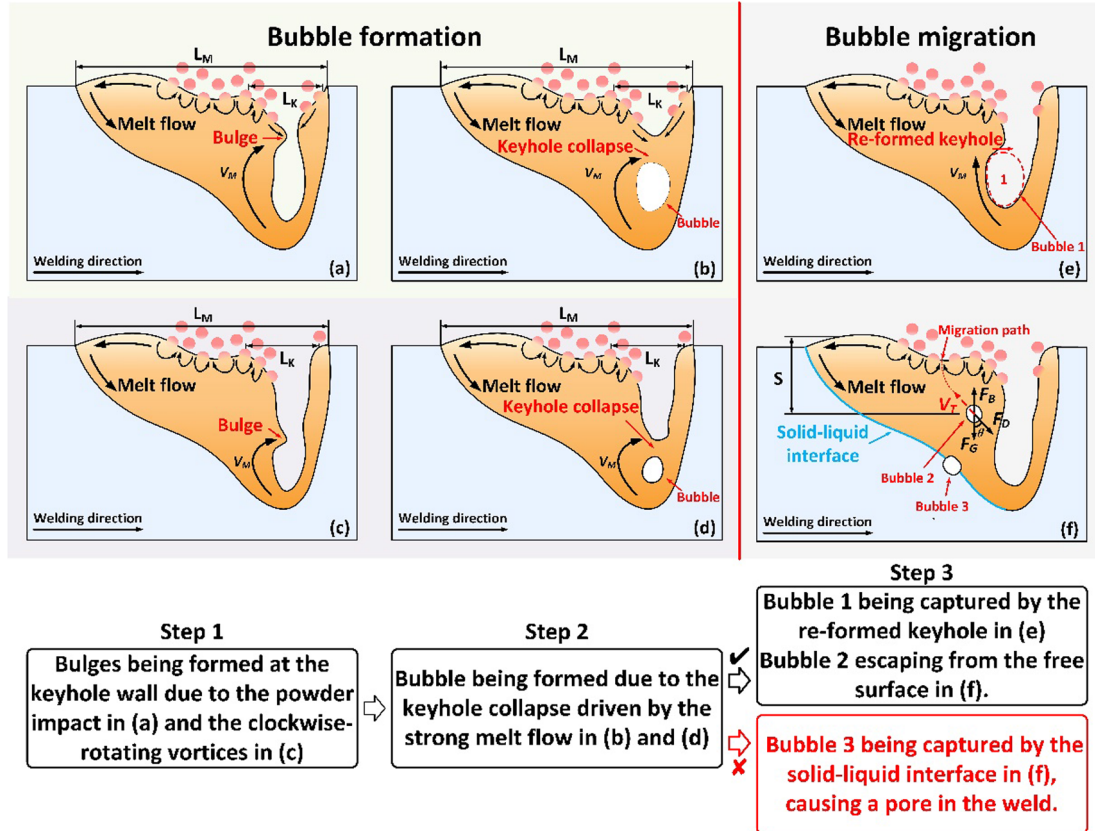


Fig. 10. Schematic diagrams of the bubble formation and migration processes: (a) bulges being formed at the keyhole wall and (b) keyhole collapse due to the powder impact, (c) bulges being formed at the keyhole wall and (d) keyhole collapse due to the clockwise-rotating vortices, (e) bubbles being captured by the re-formed keyhole, and (f) force condition, migration and capture of bubbles.

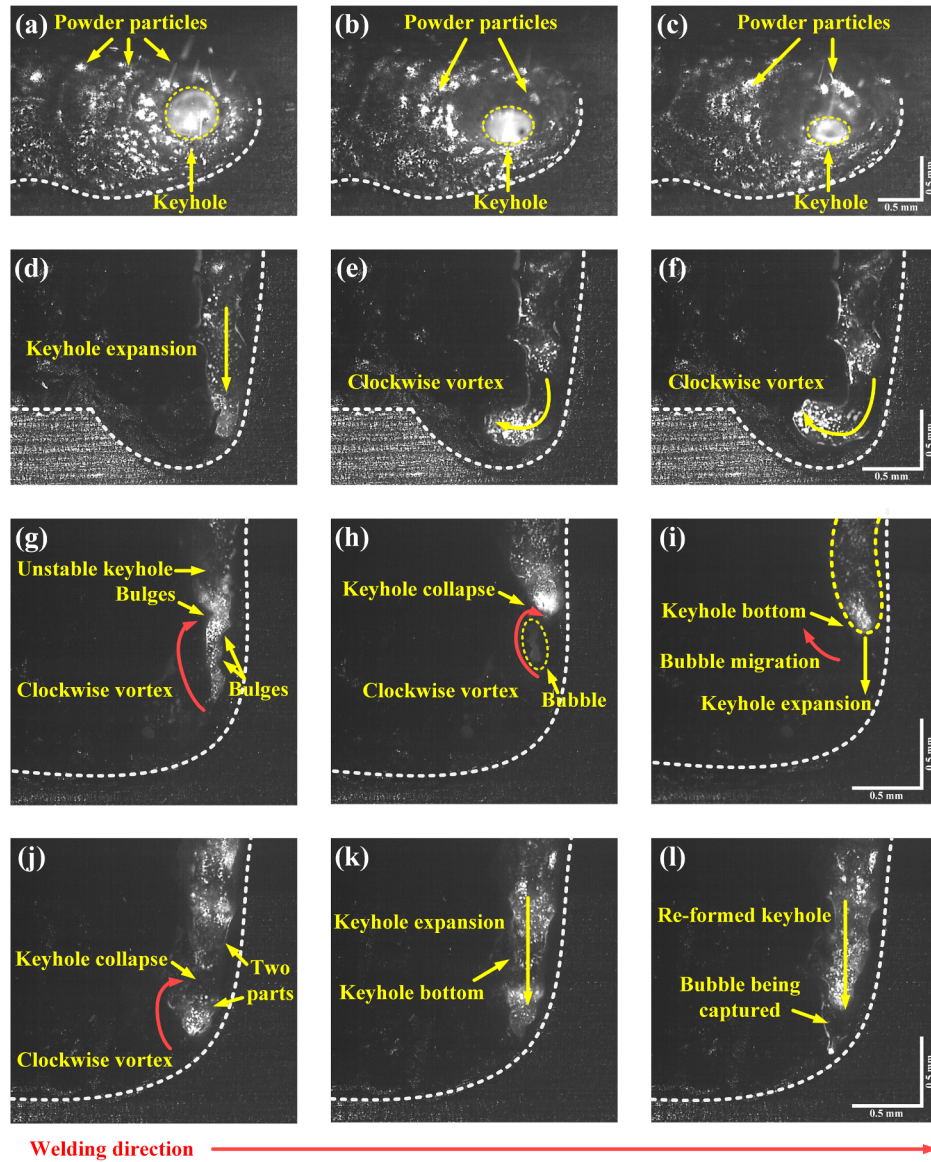


Fig. 12. Inner dynamic molten pool and keyhole behaviors with time: (a) t_1 , (b) $t_1 + 1$ ms, (c) $t_1 + 2$ ms, (d) t_2 , (e) $t_2 + 1$ ms, (f) $t_2 + 2$ ms, (g) t_3 , (h) $t_3 + 0.5$ ms, (i) $t_3 + 1$ ms, (j) t_4 , (k) $t_4 + 0.5$ ms, and (l) $t_4 + 1$ ms, where (a), (b) and (c) are corresponding to the surface molten pool behaviors, (d), (e) and (f) are corresponding to the bottom molten pool behaviors, (g), (h) and (i) show the bubble formation process, (j), (k) and (l) show the process of capturing the bubble by the re-formed keyhole.

surface molten pool behaviors corresponding to Fig. 12(a)–(c) show the decrease in keyhole entrance size when powder particles fall around the keyhole, which evidences the effects of powder particles on keyhole instability (Fig. 8). The typical clockwise vortex flow driven by the recoil pressure in the bottom of molten pool could be observed in the simulation (Fig. 5) and experiment (Fig. 12(d)–(f)). Also, the process of

bubble formation due to the strong vortex flow is found in Fig. 12(g)–(i). To be specific, Fig. 12(g) shows an unstable keyhole and a lot of bulges existing at the keyhole walls, further causing a keyhole collapse and a bubble in the bottom of keyhole under the effect of the strong melt flow (Fig. 12(h)). Meanwhile, the bubble resulting from the keyhole collapse is found in Fig. 12(j) as well. Fortunately, the bubble has no time to

Table 3

Error analysis of experimental and simulated results of the element distribution of Si.

Process	Average concentration of Si (wt%)			Error
	Metric	EV	SV	
Partial penetration welding	Overall	3.71	6.57	0.435
Full penetration welding	Upside	3.96	8.04	0.507
	Downside	2.39	4.25	0.438
	Overall	3.31	6.43	0.485

EV: Experimental values, SV: Simulated values, Error = (SV - EV)/SV.

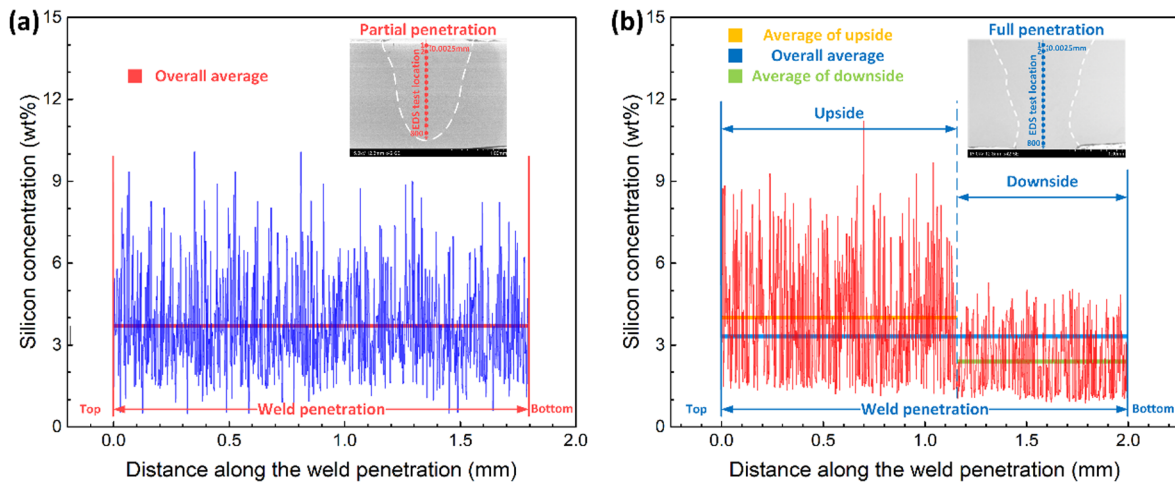


Fig. 13. Distribution of Si element along weld penetration: (a) partial penetration, (b) full penetration.

escape from the free surface due to its large size and is captured by the re-formed keyhole, leading to no pore formation as shown in Fig. 12(I). Such phenomena concerning the bubble formation and migration are consistent with the calculated results in Figs. 8 and 9.

To verify the influence of melt flow on the alloying element distribution, we compare the Si distribution along with the weld penetration in the cross sections between the partial and full penetration welding, which is characterized quantitatively by the line scan results by means of Energy Dispersive Spectrum (EDS). The error analysis of experimental and simulated results of the element distribution of Si is tabulated in Table 3. As shown in Table 3, the simulated values of Si concentration are roughly twice the experimental values considering that the diffusive effects of the alloying element are beyond consideration in our case. Despite all this, the overall trend of the alloying element distribution driven by the melt flow is the same in both experimental and simulation results. Fig. 13(a) shows that the alloying element (Si) is distributed relatively uniformly during the partial penetration welding, showing a good accordance with the simulation results in Fig. 7(a). During the full penetration welding, in agreement with the simulation results in Fig. 7(c), a higher concentration of Si is found in the upper part of molten pool, as opposed with the lower part (see Fig. 13(b)).

In summary, the proposed three-dimensional model can be used to yield a practically reasonable solution to predict the process stability and alloying element distribution under various strategies and materials during powder feeding-based laser welding, which further permits a guarantee for welding quality of manufactured products.

4. Conclusions

In this work, we develop a 3D model incorporating the powder transport, laser heating, and molten pool dynamics in an integrated manner to shed light on the role of thermal fluid dynamics on the alloying element distribution and weld porosity in PFLBW of Al alloy. The major conclusions are listed as follows:

1. The different roles of recoil pressure: driving the molten metal flow at the rear part of keyhole in partial penetration and mainly maintaining the stable keyhole profile in full penetration, are responsible for the difference between their melt flow behaviors.
2. The alloying element transport path in the molten pool in partial penetration can be generalized as the shallow areas, the rear areas, the central and bottom areas of molten pool, causing a relatively uniform alloying element distribution. However, two vortices rotating in the opposite direction in full penetration results in a higher concentration of Si in the upper part of molten pool.

3. The bubble formation and migration can be summarized in three necessary steps: bulges being formed at the keyhole wall, bubble being formed due to the keyhole collapse, and bubble being captured by the re-formed keyhole or escaping from the free surface or being captured by the solid-liquid interface to become a pore.
4. The obtained results can be used to yield a practically reasonable solution to predict the process stability and alloying element distribution under various strategies and materials during powder feeding-based laser welding, considering that the applied mathematical model has been confirmed experimentally.

Declaration of Competing Interest

We declare that we have no conflict of interest.

Acknowledgments

The authors gratefully acknowledge the financial support from the National Science and Technology Major Project of China (No. 2019ZX04025001).

References

- [1] H.J. Kim, G.A. Keoleian, S.J. Skerlos, Economic assessment of greenhouse gas emissions reduction by vehicle lightweighting using aluminum and high-strength steel, *J. Ind. Ecol.* 15 (2011) 64–80.
- [2] H.C. Chen, A.J. Pinkerton, L. Li, Z. Liu, A.T. Mistry, Gap-free fibre laser welding of Zn-coated steel on Al alloy for light-weight automotive applications, *Mater. Des.* 32 (2011) 495–504.
- [3] S.A. Tsirkas, P. Papanikos, T. Kermanidis, Numerical simulation of the laser welding process in butt-joint specimens, *J. Mater. Process. Technol.* 134 (2003) 59–69.
- [4] M. Geiger, K.H. Leitz, H. Koch, A. Otto, A 3D transient model of keyhole and melt pool dynamics in laser beam welding applied to the joining of zinc coated sheets, *Prod. Eng.* 3 (2009) 127–136.
- [5] C. Panwisawas, Y. Sovani, R.P. Turner, J.W. Brooks, H.C. Basoalto, I. Choquet, Modelling of thermal fluid dynamics for fusion welding, *J. Mater. Process. Technol.* 252 (2018) 176–182.
- [6] S. Pang, W. Chen, J. Zhou, D. Liao, Self-consistent modeling of keyhole and weld pool dynamics in tandem dual beam laser welding of aluminum alloy, *J. Mater. Process. Technol.* 217 (2015) 131–143.
- [7] F. Lu, X. Li, Z. Li, X. Tang, H. Cui, Formation and influence mechanism of keyhole-induced porosity in deep-penetration laser welding based on 3D transient modeling, *Int. J. Heat Mass Transf.* 90 (2015) 1143–1152.
- [8] Z. Lei, P. Li, X. Zhang, S. Wu, H. Zhou, L. Nannan, Microstructure and mechanical properties of welding-brazing of Ti/Al butt joints with laser melting deposition layer additive, *J. Manuf. Process.* 38 (2019) 411–421.
- [9] C. Tan, J. Yang, X. Zhao, K. Zhang, X. Song, B. Chen, L. Li, J. Feng, Influence of Ni coating on interfacial reactions and mechanical properties in laser welding-brazing of Mg/Ti butt joint, *J. Alloys Compd.* 764 (2018) 186–201.
- [10] X. Zhao, C. Tan, L. Xiao, H. Xia, B. Chen, X. Song, L. Li, J. Feng, Effect of the Ni coating thickness on laser welding-brazing of Mg/steel, *J. Alloys Compd.* 769 (2018) 1042–1058.
- [11] Z. Sun, M. Kuo, Laser welding with powder feed, *Sci. Technol. Weld. Join.* 3 (1998)

- 279–285.
- [12] S. Liang, Y. Lei, Q. Zhu, The filler powders laser welding of ODS ferritic steels, *J. Nucl. Mater.* 456 (2015) 206–210.
 - [13] M. Culleton, D. McDonnell, R. Coull, H. Shipley, G. O'Donnell, D. Trimble, R. Lupoi, Optimisation of process parameters to address fundamental challenges during selective laser melting of Ti-6Al-4V: a review, *Int. J. Mach. Tools Manuf.* 128 (2018) 1–20.
 - [14] M.F. Chiang, C. Chen, Induction-assisted laser welding of IN-738 nickel-base superalloy, *Mater. Chem. Phys.* 114 (2009) 415–419.
 - [15] B.G. Oztoprak, E. Akman, M.M. Hanon, M. Günes, S. Gümus, E. Kacar, O. Gundogdu, M. Zeren, A. Demir, Laser welding of copper with stellite 6 powder and investigation using LIBS technique, *Opt. Laser Technol.* 45 (2013) 748–755.
 - [16] H.C. Wu, L.W. Tsay, C. Chen, Laser beam welding of 2205 duplex stainless steel with metal powder additions, *Isij Int.* 44 (2004) 1720–1726.
 - [17] S. Missori, A. Sili, Structural characterization of C-Mn steel laser beam welded joints with powder filler metal, *Weld. J.* 317–323 (2000).
 - [18] N. Seto, S. Katayama, A. Matsunawa, High-speed simultaneous observation of plasma and keyhole behavior during high power CO₂ laser welding: Effect of shielding gas on porosity formation, *J. Laser Appl.* 12 (2000) 245.
 - [19] S. Katayama, Y. Naito, S. Uchiumi, M. Mizutani, Penetration, porosity prevention mechanism and welding phenomena in laser-arc hybrid welding, *Conf. Lasers Electro-Opt. Eur. - Tech. Dig.* 35 (2005) 13–18.
 - [20] S. Tsukamoto, I. Kawaguchi, G. Arakane, H. Honda, Keyhole behaviour in high power laser welding, *Spie* 4831 (2003) 251–256.
 - [21] Z. Lei, S. Wu, P. Li, B. Li, N. Lu, X. Hu, Numerical study of thermal fluid dynamics in laser welding of Al alloy with powder feeding, *Appl. Therm. Eng.* 151 (2019) 394–405.
 - [22] H. Pan, T. Sparks, Y.D. Thakar, F. Liou, The Investigation of gravity-driven metal powder flow in coaxial nozzle for laser-aided direct metal deposition process, *J. Manuf. Sci. Eng.* 128 (2006) 541.
 - [23] J.L. Arrizubieta, I. Tabernerio, J. Exequiel Ruiz, A. Lamikiz, S. Martinez, E. Ukar, Continuous coaxial nozzle design for LMD based on numerical simulation, *Phys. Proc.* 56 (2014) 429–438.
 - [24] P. Nie, O.A. Ojo, Z. Li, Modeling analysis of laser cladding of a nickel-based superalloy, *Surf. Coat. Technol.* 258 (2014) 1048–1059.
 - [25] O.B. Kovalev, I.O. Kovaleva, I.Y. Smurov, Numerical investigation of gas-disperse jet flows created by coaxial nozzles during the laser direct material deposition, *J. Mater. Process. Technol.* 249 (2017) 118–127.
 - [26] O.B. Kovalev, D.V. Bedenko, A.V. Zaitsev, Development and application of laser cladding modeling technique: from coaxial powder feeding to surface deposition and bead formation, *Appl. Math. Model.* 57 (2018) 339–359.
 - [27] S. Wen, Y.C. Shin, Modeling of transport phenomena during the coaxial laser direct deposition process, *J. Appl. Phys.* 108 (2010).
 - [28] H. Ki, J. Mazumder, P.S. Mohanty, Modeling of laser keyhole welding: Part I. mathematical modeling, numerical methodology, role of recoil pressure, multiple reflections, and free surface evolution, *Metall. Mater. Trans. A* 33 (2002) 1817–1830.
 - [29] H. Ki, P.S. Mohanty, J. Mazumder, Modelling of high-density laser-material interaction using fast level set method, *J. Phys. D. Appl. Phys.* 34 (2001) 364–372.
 - [30] C. Hirt, B. Nichols, Volume of fluid (VOF) method for the dynamics of free boundaries, *J. Comput. Phys.* 39 (1981) 201–225.
 - [31] H. Zhao, W. Niu, B. Zhang, Y. Lei, M. Kodama, T. Ishide, Modelling of keyhole dynamics and porosity formation considering the adaptive keyhole shape and three-phase coupling during deep-penetration laser welding, *J. Phys. D. Appl. Phys.* 44 (2011) 485302.
 - [32] L. Huang, X. Hua, D. Wu, F. Li, Numerical study of keyhole instability and porosity formation mechanism in laser welding of aluminum alloy and steel, *J. Mater. Process. Technol.* 252 (2018) 421–431.
 - [33] R. Lin, H. ping Wang, F. Lu, J. Solomon, B.E. Carlson, Numerical study of keyhole dynamics and keyhole-induced porosity formation in remote laser welding of Al alloys, *Int. J. Heat Mass Transf.* 108 (2017) 244–256.
 - [34] W.I. Cho, S.J. Na, C. Thomy, F. Vollertsen, Numerical simulation of molten pool dynamics in high power disk laser welding, *J. Mater. Process. Technol.* 212 (2012) 262–275.
 - [35] W.I. Cho, S.J. Na, M.H. Cho, J.S. Lee, Numerical study of alloying element distribution in CO₂ laser-GMA hybrid welding, *Comput. Mater. Sci.* 49 (2010) 792–800.
 - [36] J.H. Cho, S.J. Na, Implementation of real-time multiple reflection and Fresnel absorption of laser beam in keyhole, *J. Phys. D. Appl. Phys.* 39 (2006) 5372–5378.
 - [37] S. Muhammad, K. Miikka, N. Suck-Joo, H. Sang-Woo, K. Veli, Effect of leading and trailing torch configuration on mixing and fluid behavior of laser-gas metal arc hybrid welding, *J. Laser Appl.* 29 (2017) 042009.
 - [38] M.R.N. Esfahani, J. Coupland, S. Marimuthu, Numerical simulation of alloy composition in dissimilar laser welding, *J. Mater. Process. Technol.* 224 (2015) 135–142.
 - [39] M. Masoomi, S.M. Thompson, N. Shamsaei, Laser powder bed fusion of Ti-6Al-4V parts: thermal modeling and mechanical implications, *Int. J. Mach. Tools Manuf.* 118–119 (2017) 73–90.
 - [40] M. Xia, D. Gu, G. Yu, D. Dai, H. Chen, Q. Shi, Influence of hatch spacing on heat and mass transfer, thermodynamics and laser processability during additive manufacturing of Inconel 718 alloy, *Int. J. Mach. Tools Manuf.* 109 (2016) 147–157.
 - [41] D. Gu, M. Xia, D. Dai, On the role of powder flow behavior in fluid thermodynamics and laser processability of Ni-based composites by selective laser melting, *Int. J. Mach. Tools Manuf.* 137 (2019) 67–78.
 - [42] I.O. Kovaleva, O.B. Kovalev, Simulation of the acceleration mechanism by light-propulsion for the powder particles at laser direct material deposition, *Opt. Laser Technol.* 44 (2012) 714–725.
 - [43] I. Kovaleva, O. Kovalev, A.V. Zaitsev, D.V. Sergachev, Modeling and numerical study of light-propulsion phenomena of particles acceleration in coaxial laser powder cladding, *Phys. Proc.* 56 (2014) 439–449.
 - [44] B. Han, Y. Chen, W. Tao, H. Li, L. Li, Microstructural evolution and interfacial crack corrosion behavior of double-sided laser beam welded 2060/2099 Al-Li alloys T-joints, *Mater. Des.* 135 (2017) 353–365.
 - [45] B. Han, Y. Chen, W. Tao, Z. Lei, H. Li, S. Guo, P. Li, Nano-indentation investigation on the local softening of equiaxed zone in 2060–T8/2099-T83 aluminum-lithium alloys T-joints welded by double-sided laser beam welding, *J. Alloys Compd.* 756 (2018) 145–162.
 - [46] D. Fuster, A. Bagué, T. Boeck, L. Le Moyne, A. Leboissetier, S. Popinet, P. Ray, R. Scardovelli, S. Zaleski, Simulation of primary atomization with an octree adaptive mesh refinement and VOF method, *Int. J. Multiph. Flow.* 35 (2009) 550–565.
 - [47] J. Brackbill, D. Kothe, C. Zemach, A continuum method for modeling surface tension, *J. Comput. Phys.* 100 (1992) 335–354.
 - [48] R. Hu, X. Chen, G. Yang, S. Gong, S. Pang, Metal transfer in wire feeding-based electron beam 3D printing: Modes, dynamics, and transition criterion, *Int. J. Heat Mass Transf.* 126 (2018) 877–887.
 - [49] X. Li, F. Lu, H. Cui, X. Tang, Y. Wu, Numerical modeling on the formation process of keyhole-induced porosity for laser welding steel with T-joint, *Int. J. Adv. Manuf. Technol.* 72 (2014) 241–254.
 - [50] C. Zhang, M. Gao, D. Wang, J. Yin, X. Zeng, Relationship between pool characteristic and weld porosity in laser arc hybrid welding of AA6082 aluminum alloy, *J. Mater. Process. Technol.* 240 (2017) 217–222.